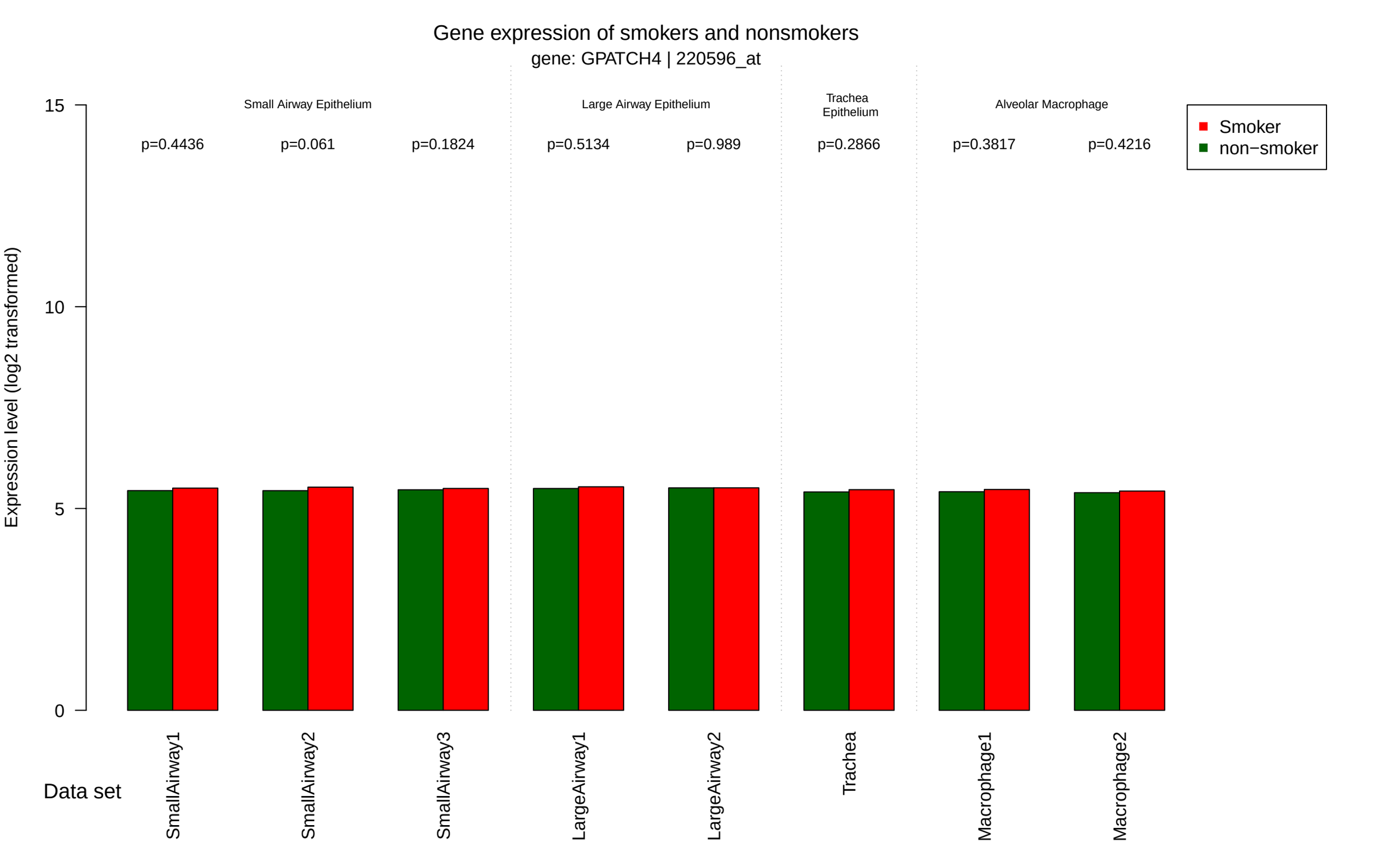
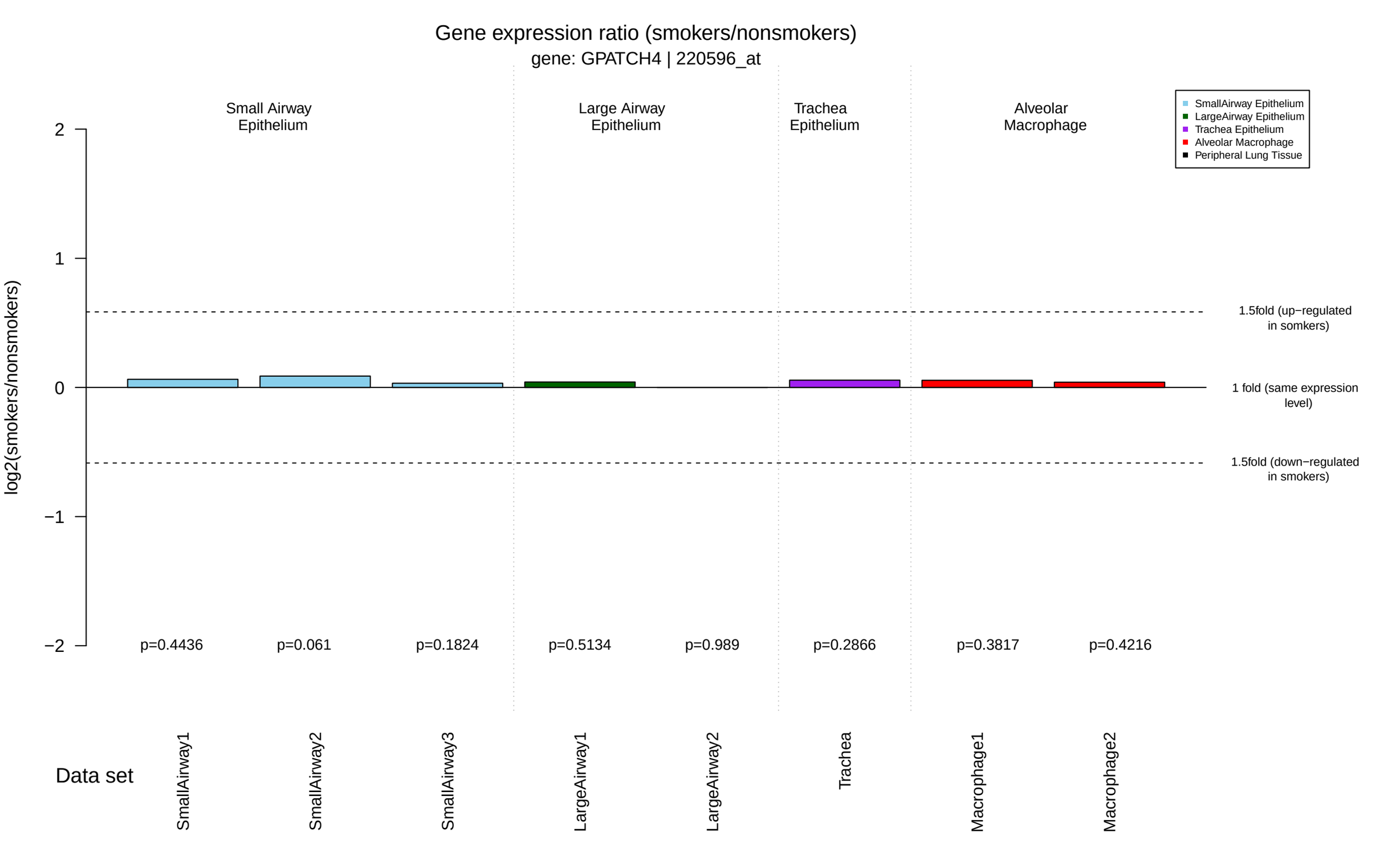
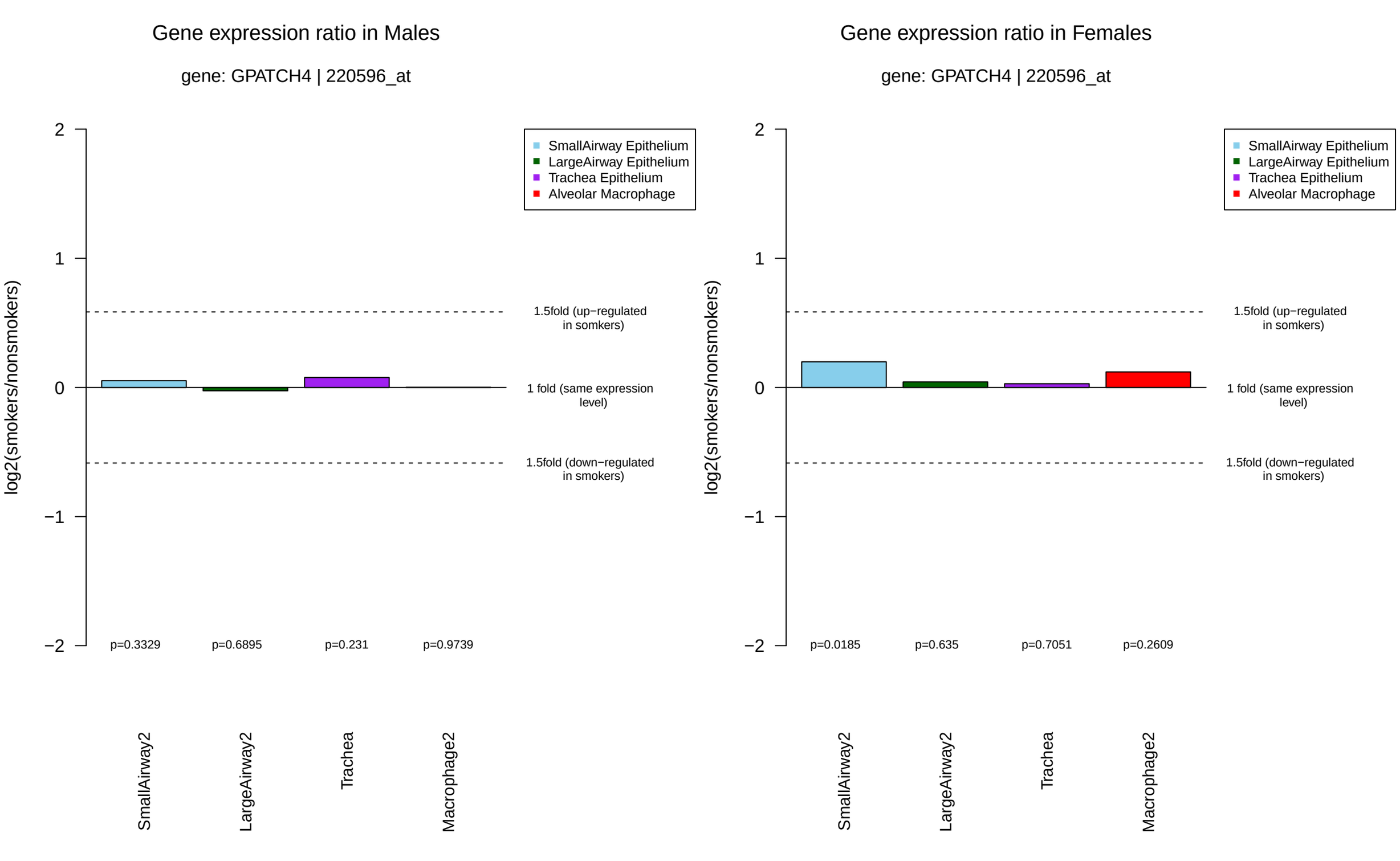
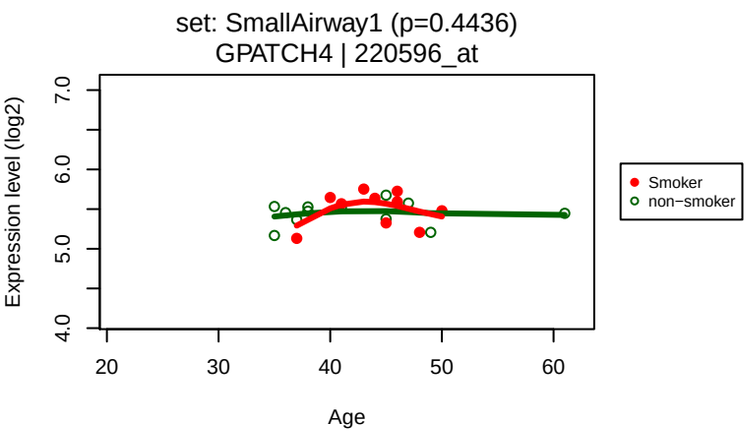
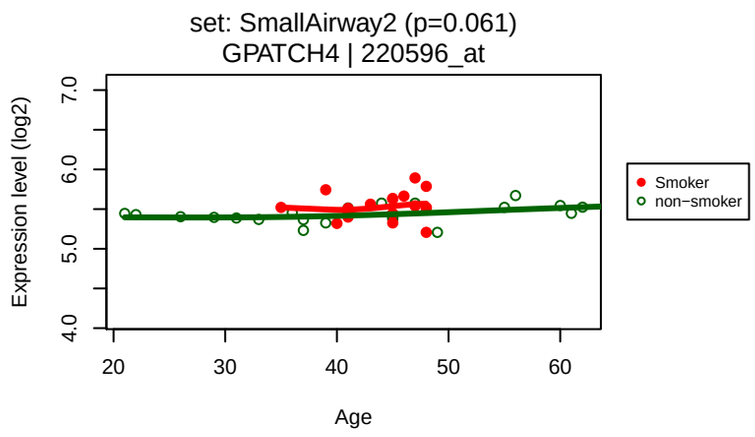
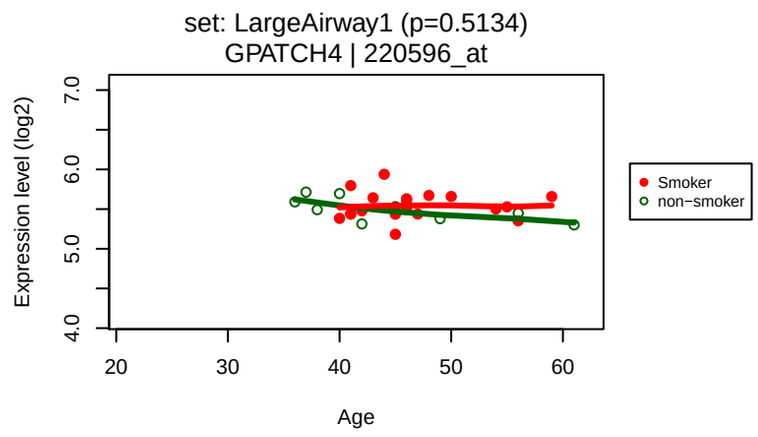

Gene expression of smokers and nonsmokers
gene: GPATCH4 | 220596_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4436
p=0.061
p=0.1824
p=0.5134
p=0.989
p=0.2866
p=0.3817
p=0.4216
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

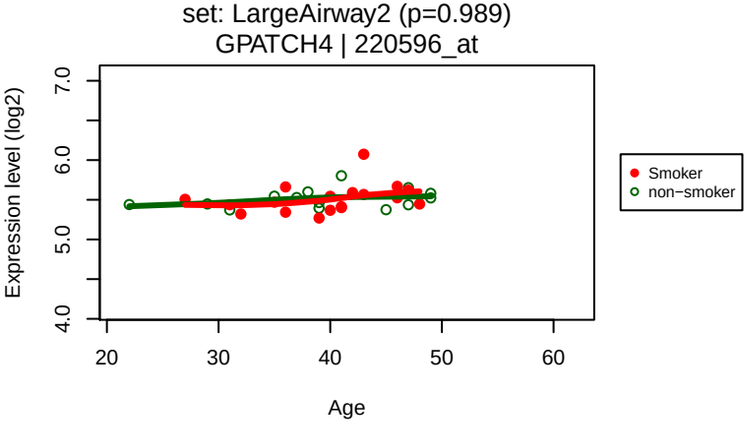

Gene expression ratio (smokers/nonsmokers)
gene: GPATCH4 | 220596_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4436
p=0.061
p=0.1824
p=0.5134
p=0.989
p=0.2866
p=0.3817
p=0.4216
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

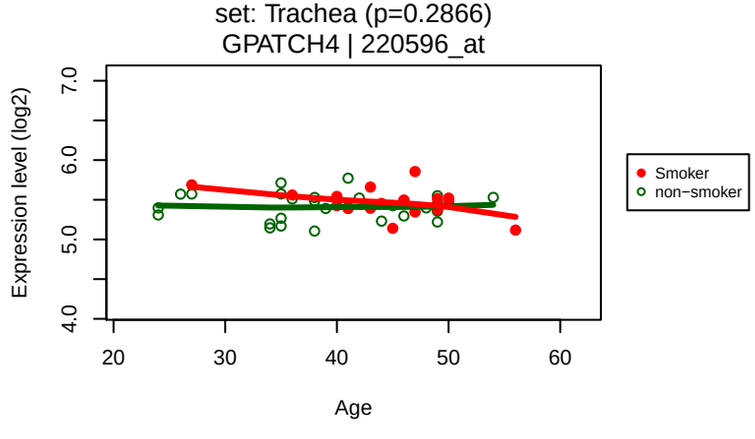

Gene expression ratio in Males
gene: GPATCH4 | 220596_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3329
p=0.6895
p=0.231
p=0.9739
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPATCH4 | 220596_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0185
p=0.635
p=0.7051
p=0.2609
SmallAirway2
LargeAirway2
Trachea
Macrophage2

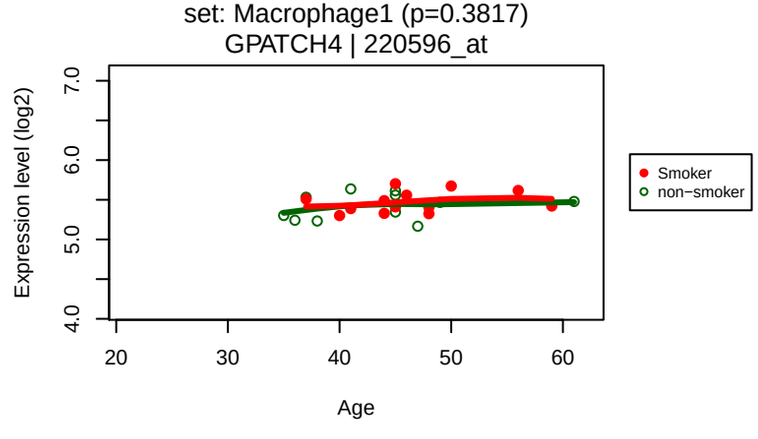

set: SmallAirway1 (p=0.4436)
GPATCH4 | 220596_at
Expression level (log2)
Age
Smoker
non−smoker

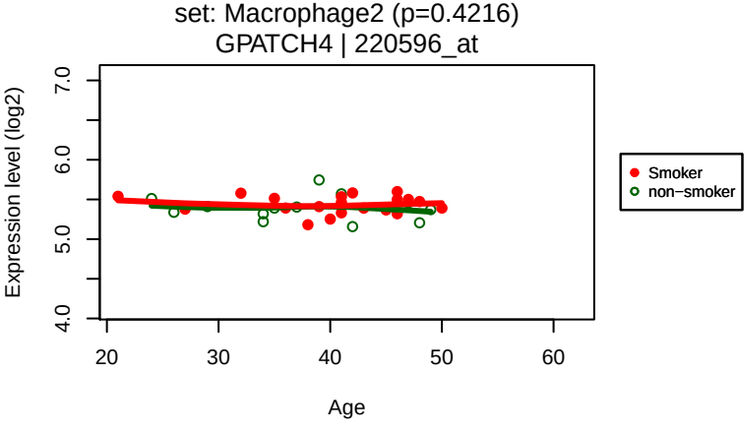

set: SmallAirway2 (p=0.061)
GPATCH4 | 220596_at
Expression level (log2)
Age
Smoker
non−smoker

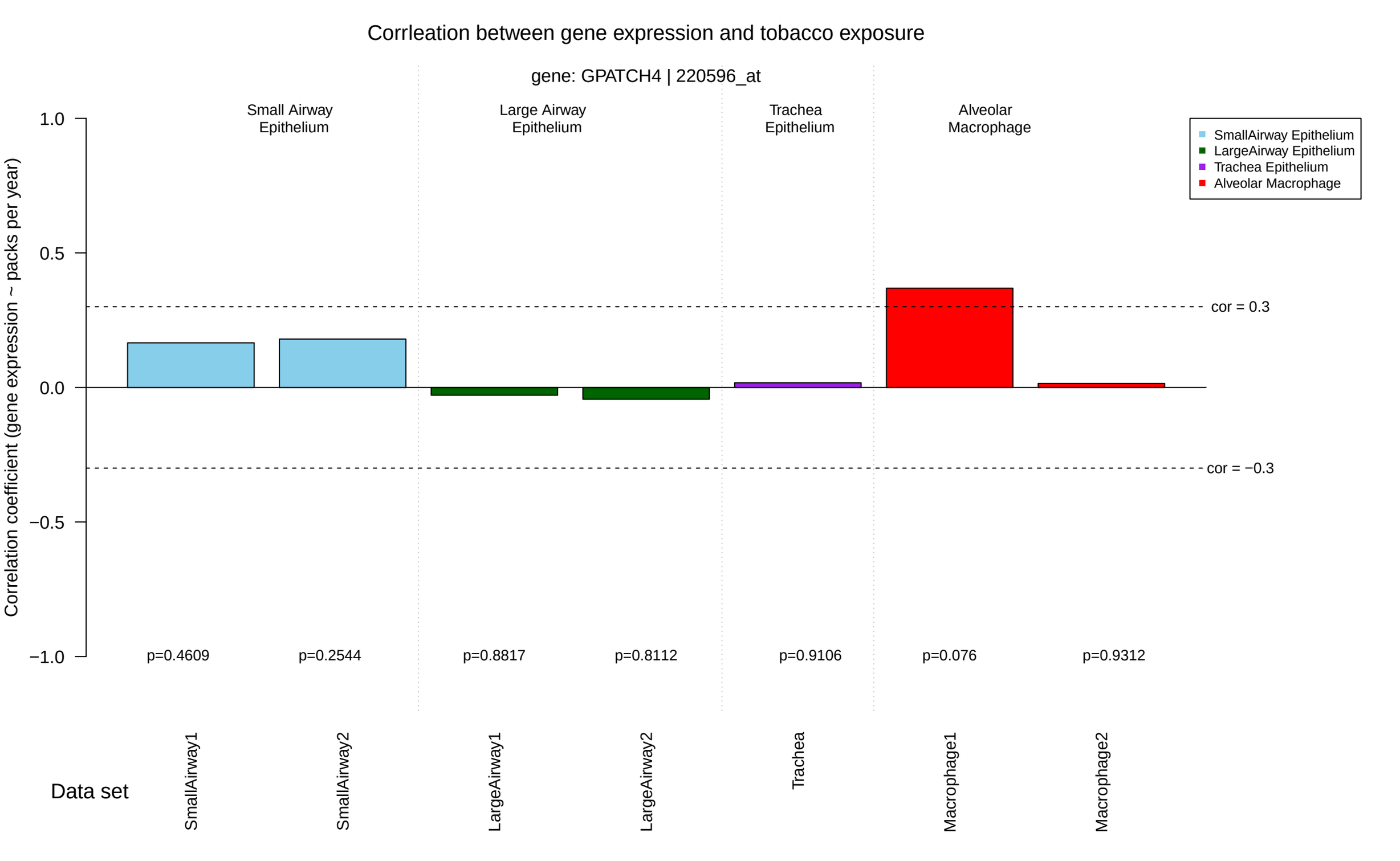

set: LargeAirway1 (p=0.5134)
GPATCH4 | 220596_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.989)
GPATCH4 | 220596_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2866)
GPATCH4 | 220596_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3817)
GPATCH4 | 220596_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4216)
GPATCH4 | 220596_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPATCH4 | 220596_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4609
p=0.2544
p=0.8817
p=0.8112
p=0.9106
p=0.076
p=0.9312
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2